

Walkthrough: Image to 3D Textured Mesh

This guide explains the entire 6-step pipeline from loading a 2D image generating a textured 3D model.

■ Step 1: Image Basics

- **Purpose:** Verify the environment and image loading.
 - **Script:** `01_image_basics.py`
 - **Action:** Opens an image using OpenCV and Pillow to ensure dependencies are installed.
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■■ Step 2: Image Processing

- **Purpose:** Prepare the image for AI.
 - **Script:** `02_image_processing.py`
 - **Action:**
 1. Resizes images to fit standard input sizes (e.g., 512x512).
 2. Converts colors from BGR (OpenCV) to RGB.
 3. Outputs a "Processed" version in `output/processed.jpg`.
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■ Step 3: Depth Estimation (The Core AI)

- **Purpose:** Predict the "depth" (distance) of every pixel.
 - **Script:** `03_depth_estimation.py`
 - **Technology:**
 - * **Model:** [Depth Anything V2 (Small)](<https://huggingface.co/depth-anything/Depth-Anything-V2-Small>)
 - * **Refinement:** Uses a **Guided Filter** to sharpen depth edges using the original image as a guide.
 - **Output:**
 1. `point_cloud_rgb.ply` (3D Points with Color)
 2. `depth_map.npy` (Raw depth data for later steps)
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■■ Step 4: Visualize Point Cloud

- **Purpose:** Inspect the raw 3D data.
 - **Script:** `04_visualize_point_cloud.py`
 - **Action:** Opens an interactive 3D viewer (Open3D) to rotate/zoom the point cloud.
 - **Look For:** Verify the depth looks correct (objects popping out).
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■ Step 5: Geometry Mesh (Shape Only)

- **Purpose:** Convert separate points into a solid surface.
 - **Script:** 05_mesh_generation.py
 - **Method: Poisson Surface Reconstruction.**
 - **Action:** mathematically wraps a "skin" around the points.
 - **Output:** model_mesh_geometry.obj (Solid white shape).
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■ Step 6: Textured Mesh (Final Result)

- **Purpose:** Apply the high-resolution photo onto the 3D shape.
 - **Script:** 06_mesh_textured.py
 - **Method: Displacement Mapping.**
1. Creates a 3D grid based on the depth map.
 2. Maps the original image as a UV Texture.
 3. Optimizes geometry for low-end devices (CPU friendly).
- **Output:** final_model/model.obj + model.mtl + Texture.
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■ How to View the Result

The final output is a standard **OBJ** file. To view it with textures:

1. Open the final_model folder.
 2. Select **ALL 3 FILES** (model.obj, model.mtl, and the .jpg).
 3. Drag them together into [3DViewer.net](https://3dviewer.net/).
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■■ Known Limitations

1. **2.5D Only:** This method creates a "relief" (like a mask). It cannot verify what is behind the object.
2. **Stretching:** The top/sides of objects may look stretched because the AI has to guess the pixels there.